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S 401 DESIGN AND FABRICATION OF STRUCTURAL STEEL

1. SCOPE OF WORK

1.1 The scope of work shall be in accordance with Part 4, Subsections 1.2 “Summary of Work” and S-001 “General Design Requirement of Structure, Civil and Building Works”.

1.2 Design and fabrication of structural steel, as defined in the AISC Code of Standards Practice, shall conform to the requirements of this Section, and all applicable requirements of the AISC Manual of Steel Construction, latest edition unless otherwise indicated. The work shall include the following :

Designing, furnishing, fabricating and delivering structural steel and crane rails including floor loading signs.

1.2.1 Designing, furnishing, fabricating and delivering gallery and auxiliary steel.

1.2.2 Designing and furnishing shop and field connections included in the structural steel.

1.2.3 Verifying field measurements for fabrication as required.

1.3 Intent Of Specification

The work covered by the specification shall include full provision for the following:

1.3.1 All structural steel, columns, girders, beams, bracing, base plates, column anchor bolts, trolley beams and other materials as required for the support of the buildings, structures, ducts, equipment, piping and other appurtenances above the foundations at grade.

1.3.2 With respect to the Bid Drawings included in the Bid Document, the following shall be clearly understood:

1.3.2.1 Work indicated on those drawings shall be used only as a guide for the type of steel work which will be required for the work; actual sizes, weights, thickness, relative quantities of various classes etc. shall be based on Bidder’s detailed design drawings.

- 1.3.2.2 Cost incurred as a result of changes due to the Bidder's design need, misinterpretation of design intent, miscalculation of connections or other design, detailing and fabrication errors shall be the Bidder's responsibility.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 'Codes and Standards'.
- 2.2 The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 'Codes and Standards' are specifically applicable to the design, manufacture and testing of the work included in this specification:

JIS		Japanese Industrial Standards
AISC		American Institute of Steel Construction
ANSI		American National Standards Institute
AREA		American Railway Engineering Association
ASTM		American society for Testing and Materials
AWS		American Welding Society
SSPC		Steel Structures Painting Council, USA
UBC		Uniform Building Code of USA
JEC		Japan Electrotechnical Committee

3. REPORT AND CERTIFICATES

Copies of material certifications, mill test report, and other report required by applicable codes or standards shall be available from the Bidder. Those reports shall be submitted as requested by the Owner after completion of the required tests.

4. MATERIALS

All materials for the power plant buildings and other structures shall conform to the requirements of the Bidder's design drawings, JIS, ASTM, Table S-401-1 and S-401-2 or approved equivalent.

Hot rolled steel of wide flange shapes with 600 mm or less depth shall be of local origin as long as available.

Table S-401-1 MATERIAL FOR STRUCTURAL STEEL WORK

Item No.	MATERIAL	ASTM STANDARDS and CODES JIS UNLESS NOTED
1.	Hot rolled beams, bars channels, plates and angles, bracing, structural tees	A36, JIS G 3101 or equivalent
2.	Plate girders	A36, JIS G 3101 or equivalent A572 Grade 50, JIS G 3136 or equivalent A588 Grade 50, JIS G 3114 or equivalent A588 Grade 46, JIS G 3114 or equivalent A588 Grade 42, JIS G 3114 or equivalent
3.	Columns	A36 or A572 or A588 or, JIS G3101 or G3136 or G3114 or Equivalent
4.	Cover plates	A36 or JIS G 3101 or A572 Grade 50 or JIS G3136 or A588 Grade 50 or equivalent
5.	Column base plates and splice plates	A36, JIS G 3101 or equivalent
6.	Connection bolts	A325F, JIS B 1186, JSSII-09 or equivalent
7.	Welding	AWS D 1.1
8.	Column anchor bolt	A36, or A307 or A193 Grade B7 or JIS G 3101 or JIS B 1178 or equivalent
9.	Round tubular bracing	A53, JIS G 3452
10.	Square and rectangular tubing	A500, Grade B or A501 or JIS G3444 or JIS G 3466
11.	Castings	A27, Grade 65-35 or A148, Grade 80-50
12.	Forgings	A668, Classes C, E and F

Note: For additional requirements for material see Table S-401-2

Table S-401-2 ADDITIONAL REQUIREMENTS FOR STRUCTURAL STEEL MATERIALS

Material Item	Standard	Additional requirements
1. Structural	ASTM A36, JIS G3101 or equivalent	Use for all structural steel other than steels specifically indicated in this specifications
2. High-strength Steel	a. ASTM A572, JIS G3136 or equivalent b. ASTM A588, JIS G3144 or equivalent	Grade, as required Use yield, as required
3. Low-sulfur High-Strength Steel (for plates over 12mm. in thickness)	ASTM A588 Grade A or B A572 Grade 50 with special process (such as vacuum degassing)	
4. Stainless Steel: Hot-rolled, annealed and pickled (HRAP) finish	a. ASTM A240 b. ASTM A240 conforming to A358, Class 1, Grade 304	Plates Plates for fabrication of welded plate Piping included in structural steelwork
5. Tubular Members : 5.1 Round Tubular Bracing	ASTM A53 of the following type and grade : a. Type E or S, open hearth or basic oxygen steel only, with sulphur content not exceeding b. Grade B only (2,460 kg/cm ² yield)	Use standard, extra strong or double extra strong weight
5.2. Rectangular or square Tubing	ASTM A500 Grade B or A501	For tubular framing

Material Item	Standard	Additional requirements
6. High-Strength Bolts	AISC Specification for Structural Joints using ASTM A325	a. Bolts and Nuts : Use the following material unless otherwise indicated. a.11. For bolts sized use A325, type 1 steel. b. Washer: A325 Bolts: One washer under the head or nut, whichever is the element turned in tightening.
7. Shoulder bolts for slotted sliding connection. 7.1 Bolts	ASTM A325, TYPE 1	22 mm diameter shank; grip length of bolt shall equal the thickness of the mated parts plus two 3 mm thick washers plus 3 mm. bolts threads shall be 19 mm diameter and the length shall equal the thickness of two nuts plus 6 mm.
7.2 Nuts	ASTM A563, grade C	Two required per bolt.
7.3 Washer	ASTM F436	Two 22 mm diameter washers required. One washer under the bolt head and one under the nut.
8. Castings	a. ASTM A27, Grade 65-35 b. ASTM A148, Grade 80-50	Use for all castings unless otherwise indicated Use only where high strength steel casting are required.
9. Forgings	ASTM A688, Classes C, E and F	Use for all forgings unless otherwise indicated

Material Item	Standard	Additional Requirements
10. Stock Material	As specified in items 1 through 10	
11. Crane Rails/weight 50 kg/m or more. (lighter rails shall conform to manufacturers Standards).	ASTM A1, No.1 Rails , with carbon content 0,67-0,82% or manufacturers standards meeting the same requirements.	a. Length: standard 12 m, subject tolerances of ASTM A1 b. Size: as indicated on the design drawings. c. End Drilling: Three holes each end, spaced to suit specified joints bars and as required for tight end joints. d. Marking: As specified in ASTM A1.
12. Crane Rail Accessories	As indicated	
12.1 Joint Bars	ASTM A3	
12.2 Bolts and Nuts	ASTM B325 and ANSI B18.2.	
12.3 Spring Washers	AREA specifications for spring washers.	Alloy steel with one washer for each bolt.
12.4 Rail Clips	ASTM A 36 or better	Shall be "Tight Fit" type, made of pressed steel plate with reversible steel fillers and 2 holes each clip.
12.5 Resilient Pad		The resilient pad under the crane rails shall be manufactured from synthetic elastomer and be resistant to wear, shear, crushing, oil, grease, oxygen and ultra-violet rays and shall be reinforced along its length with a steel strip.

5. DESIGN

5.1 General Provisions

- 5.1.1 The design of all structural steel required by this Contract and the design work for all shop and field connections, including angles, welds, web reinforcement for coped beams, gusset plates and bolts shall be performed by the Bidder. The design and shop drawings including typical calculations for all structural members and each connection type shall be submitted for review and approval by the Owner prior to fabrication. Such review and approval shall in no way be interpreted as constituting approval of the adequacy of such design for their intended use and shall not relieve the Bidder from correctness and of complete design responsibility and from fulfilling all obligations under the Contract.
- 5.1.2 All design calculations and drawings shall be prepared under the supervision of a qualified Engineer and submit to the Owner for approval prior to commencement of the fabrication.

5.2 Design Requirements

- 5.2.1 The design of structural steel shall conform to all applicable requirements of the AISC Manual of Steel Construction, latest edition and also conform to the requirements of the following specifications:
- 5.2.1.1 “Welding in Steel Structures” Part 4, Specification S-410
- 5.2.1.2 “Erection of Structural Steel” Part 4, Specifications S-402
- 5.2.2 The connections shall be designed to resist ultimate moments, shears and axial forces of connected members.
- 5.2.3 Prying action shall be considered in the design of bolted connections in accordance with the AISC Manual of Steel Construction.
- 5.2.4 The following allowable stresses specified in the AISC specification for Structural joints Using ASTM A325 bolts shall be used in connection bolt design:
- 5.2.4.1 Allowable stresses for blast-cleaned surfaces (for shop welded and field bolted connections).

- 5.2.4.2 Allowable stresses for mill scale surfaces (for shop bolted and field bolted connections).

6. FABRICATION

6.1 General Provisions

6.1.1 Conformance

Fabrication of structural steel shall conform to all applicable requirements of codes and standards and other standard specifications specified in this section, and to all requirements of this Clause 6.

6.1.2 Fabrication requirements for plumbing up and leveling

- 6.1.2.1 The structures shall be plumbed and leveled such that deviations in the completed work do not exceed the requirements of the AISC Code of Standard Practice.

- 6.1.2.2 These requirements shall be fully considered in fabrication, and Bidder's shop erection drawings shall be clearly marked with these restrictions.

6.1.3 Tolerance

The tolerances on cross sectional dimensions of rolled sections after fabrication shall be as specified in ASTM A36 or JIS G3126 or equivalent unless otherwise specified.

- 6.1.3.1 A structural member before erection shall not deviate from straightness (or the specified shape) by more than the following:

- a. For compression members : 1/1000 of the length between points that are to be laterally restrained.
- b. For other members : 1/500 at the overall length, but not more than 25 mm, unless otherwise specified. The length of a member shall not deviate from its length by more than the following :
- c. For compression members faced at both ends for bearings : ± 1 mm
- d. For other members : $+ 0\text{mm} - 4$ mm
- e. Machine surface within : ± 0.025 mm

6.1.3.2 For members as trusses and lattice girders, these tolerances apply to the member as a whole. The length of component parts shall be such that the member can properly assembled with the required accuracy.

6.1.3.3 Where two steel surfaces are required to be in contact with bearing, the maximum clearance between the bearing surfaces shall not exceed 1 mm when the members in contact are aligned within 1/1000 of their combined overall length.

6.1.4 Misfits

6.1.4.1 Attention is specifically directed to the requirements of the AISC Code of Standards Practice and Quality Criteria and Inspection Standards, with respect to correction of fabrications errors. Any errors in shop work which prevent proper field assembly and fit up will be reported, as soon as they are discovered to the Owner and to the fabricator by the Bidder. The Bidder shall then make immediate substitutes or arrange for correction at no cost to the Owner.

6.1.4.2 The allowable limits for reaming the connection bolts holes for fit up in the shop or in the field shall conform to the requirements of the AISC Quality Criteria and Inspection Standards. If fit up of bolts during erection in the field cannot be obtained by reaming the fabricated bolts holes within these limits, it is considered that there are errors in fabrication, and these errors shall be rectified in accordance with Clause 6.1.3.1.

6.2 Connections

All connections shall be welded or high strength bolted, in the shop or field as follows:

6.2.1 Shop Connections

6.2.1.1 All members and details shall be fabricated by welding or high strength bolting, using TORQUE SHEAR / FRICTION TYPE connections, and welding may be used in combination with the high strength bolting completed prior to welding.

6.2.1.2 The Bidder shall submit with his Bid Proposal, a schedule of where he proposes to use each method of fabrication, and in general the members, connections, etc. included under each method.

6.2.2 Field Connections

All connections shall be made with high-strength bolts, using FRICTION TYPE connections, unless otherwise indicated.

6.2.3 Welding

6.2.3.1 Conformance

Welding, including stud welding, shall conform to the requirements of Specification S-410 "Welding in Steel Structures".

6.2.3.2 Welding Drawing

All welding shall be clearly indicated on the shop drawings.

6.2.3.3 Field Welding Preparations

The Bidder shall provide all end and edge preparations for field welding.

6.2.4 High Strength Bolting

6.2.4.1 Conformance

High strength bolting shall conform to the applicable requirements of the AISC Specification for Structural Joints Using ASTM A325 Bolts or JIS B1186 or approved equivalent for friction type connections, and to the requirements of this Clause 6.2.4 unless otherwise indicated.

6.2.4.2 Bolt Types

Shall be as indicated in Table S-401-2.

6.2.4.3 Bolt holes

Shall be 1.5 mm larger than the nominal diameter of the bolt. Larger or slot-
ted holes will not be permitted unless otherwise indicated.

6.2.4.4 Long Grips

Additional high-strength bolts are not required for long grips.

6.2.4.5 Installations

- a. Torque shear type high strength bolt (TC bolt) specified in ASTM A325 shall be used in principle.
- b. The turn-of-nut tightening or calibrated wrench tightening methods, as specified in the JIS or AISC specifications for structural Joints Using ASTM 325 Bolts, shall be used for sundry steel and galvanized steel. The use of a direct tension indicator for tightening will not be permitted.
- c. The minimum projection of the installed bolt beyond the nut shall be 6 mm.
- d. The threaded part of the installed bolt shall not be in the shear plane.
- e. Reuse of torque bolts will not be permitted.

6.2.5 Beam Connections

6.2.5.1 Connection types

Shall conform to the AISC Specification, part 4.

6.2.5.2 Connection Angles

- a. Connection angles shall be minimum 10 mm thick
- b. Connection angles for any deep girder shall be continuous from the top flange to the bottom flange in all cases.
- c. Usual gauge as presented by AISC shall be used. Bolt staggering shall be used only to maintain the usual gauge.

6.2.5.3 Knife Connection

This connection, consisting of two connection angles shop assemble to receive the web of a beam in the field erection, shall not be used.

6.3 Miscellaneous Details

6.3.1 Welded Cover Plates on Columns and Girders

Cover plates on each flange of columns or girders shall be a single plate in each thickness.

6.3.2 Bracing

Vertical diagonal bracing members shall be of wide flange shapes, double angles, double channels or tubular.

Horizontal bracing shall be single angles.

6.3.2.1 Tubular Bracing

- a. Tubular bracing used for indoor work (not exposed to the elements) shall be closed or capped.
- b. Tubular bracing used for exterior work (exposed to the elements) shall have the top end capped and the bottom end open. Cap plates shall be 6 mm thick and continuously welded all around the bracing.

6.3.2.2 Draw for Single-Angles Cross Bracing

Long single-angles cross bracing members shall be fabricated with a draw or reduction in the actual distance centre-to-centre of end connection bolt holes (as compared to the theoretically correct distance) as indicated in following Table S-401-3; no draw shall be provided for double-angle, tubular or other shape bracing:

Table S-401-3 DRAW REQUIREMENTS FOR SINGLE ANGLE BRACING

Length	Draw
0 to 3.5 m	-
Over 3.5 m to 7 m	1.5 mm
Over 7 m to 10 m	3 mm
Over 10 m	4.5 mm

6.3.2.3 Double-Angle Bracing

All double-angle bracing members shall have long legs back-to-back unless otherwise required.

6.3.2.4 Gusset Plates

Gusset plates for horizontal and vertical bracing shall be a minimum of 6 mm and 9 mm thick respectively.

6.3.3 Camber

Built-up girders, trusses and rolled beams shall be fabricated to provide for camber in accordance with the requirements of the AISC Manual of Steel Construction. 9 mm thick respectively.

6.3.4 Column Splice Milled Ends

These ends shall not be out of square more than 0.75 mm, as specified in ASTM A6 Table 24. However, only milling techniques shall be used in this fabrication.

6.3.5 Curb Angles and Guard Plates

6.3.5.1 Curb angles and guard plates shall be connected to the structural steel in the shop.

6.3.5.2 Guard plates shall be furnished in as large lengths as possible, with joints not closer than 90 cm to handrail posts. Cutouts where required to clear indicated piping, or other equipment, etc., shall be provided.

6.3.5.3 For adjoining guard plates in continuous runs or at corners, the ends shall be tack-welded together at the top.

6.3.6 Frames for Removable Concrete Slabs

Angle frames shall be furnished for removable concrete slabs supported on structural steel framing. The Bidder will provide lifting inserts for these slabs. Angles shall have headed studs for sound anchorage in concrete.

6.3.7 Hangers

All hangers shall have a minimum of two high strength bolts at each end.

6.3.8 Trolley Beam Stops

Each trolley beam stop shall consist of two 75 mm x 75 mm x 12 mm angles, each of which shall be connected to the beam with two 18 mm diameter machine bolts. Each bolt shall have hexagon head and one heavy hexagon nut.

6.3.9 Girts

6.3.9.1 Extent

Girts shall include horizontal girts, carrier girts, girt posts, door and window framing attached to girts, parapet framing, sag rods etc.

6.3.9.2 Plumb and Level

Girts shall be fabricated to the same plumb and level requirements as specified in Clause 6.1.2 foregoing.

6.3.9.3 Frames Around Openings

Girts around doors, windows, and louver openings shall be fabricated in accordance with the following requirements:

- a. All corners of openings formed by junctions of vertical and horizontal girt members shall be absolutely square.
- b. Vertical and horizontal dimensions of openings shall each be within an overall tolerance of ± 1.5 mm for doors, ± 6 mm for windows, to 0 mm for louvers.
- c. The junctions of vertical and horizontal girt members at the head and sill of door openings shall be detailed with end clearance not exceeding 6 mm nor less than 3 mm, and shall be arranged for field welding closures, of these junctions.
- d. Head angles for windows shall be detailed with end clearances not exceeding 6 mm, nor less than 3 mm; these clearances shall apply for intermediate joints in continuous runs of head angles, and also for joints of junctions of the head angle and jamb angle. Also, these joints shall be detailed for field welding closures.

- e. Fill plates at the junction of jamb angles and sill angles shall be provided, and shall be detailed for field welding closures of these junctions.
- f. Steel channel door frames may be shipped knocked-down for field welded assembly or may be completely shop-assembled and shipped as a unit.

6.3.9.4 Connections

Each connection for a girt shall have a minimum of two 16 mm diameter high-strength bolts unless otherwise indicated.

6.3.9.5 Sag Rods

All sag rods shall have two heavy hexagon nuts each end. Holes for sag rods shall be 1.5 mm larger than the nominal diameter of the sag rods. Larger holes up to 4.5 mm larger than the rod diameter will be permitted only if hardened washer is used with each nut on each side of these holes.

6.3.10 Crane Rails

6.3.10.1 Rails lengths shall be fabricated so that rail joints on opposite sides of the runway are staggered with respect to each other. Rail joints shall not coincide with the joints occurring in the runway support girders.

6.3.10.2 Rails shall be milled or ground grind to provide tight joint assembly.

6.3.10.3 Rail clips shall be Tight type in pairs spaced at 60 cm on centers.

6.4 Shop Cleaning And Painting

6.4.1 Shop cleaning and painting including surface preparation of structural steel shall conform to all applicable requirements of specification S-420 “ Prime Coat and Shop Finish Painting”.

6.4.2 Identification of Trolley Beams

All trolley beam surfaces shall be painted yellow with one shop prime coat of paint. The design capacity of each trolley beam shall be stenciled on both sides of the web at the approximate midspan of each beam, using the notation “CAPACITY _____ TONS” lettered in 75 mm high block letters with

black paint. The surface preparation, painting, and paint products/manufacturers shall conform to the applicable requirements of S-420.

6.4.3 Correction of Painting Work

6.4.3.1 Painted surfaces of structural steel will be inspected, as specified in specification S-420 “Prime Coat and Shop Finish Painting” in the field after a reasonable period of weathering. If the condition of any surface indicates insufficient thickness of paint was used or method of application was incorrect, etc. (regardless of prior approval of the work in the shop), the Bidder shall repaint all such surfaces in the field as requested.

6.4.3.2 The foregoing requirements shall not apply with respect to surfaces abraded in shipping, handling etc. for which field touchup painting will be performed by the Bidder.

7. SWITCHYARD STEEL STRUCTURE

7.1 General

7.1.1 Steel structures for supporting the insulators, switchgear, overhead conductors, bus-bars, earth-wires, and other equipment shall be provided under this Contract as shown on the bidding drawings. And they have to be designed and erected in accordance with the relevant internationally recognized standards. The Bidder’s design and drawings shall be submitted to the Owner for approval.

7.1.2 Existing substations to be extended have gantry steel structures, a price shall be included for a design closely resembling the existing arrangement of the Bidder's manufacturer. Drawings of the Bidder's design are to be submitted to the Owner for approval.

7.1.3 The structures shall include all necessary access ladders to give access to the various levels of the high-level equipment and shall incorporate all necessary screens to comply with the requirements of insulation levels and minimum clearances.

7.1.4 The design and arrangement of supporting structures shall be subject to approval by the Owner. The structures shall be rigid and self-bracing against all dead, wind, pull-off and other applied loads. Wherever such an arrange-

ment can be adopted, structures shall be braced by horizontal beams at intermediate or high level to provide an integrated framework. At or near ground level, all uprights shall be provided with holding down bolts provided under this Contract.

- 7.1.5 The rigidity of the structures shall be such that the alignment of the apparatus which they carry shall not be disturbed by the loads to which the structures are subjected. Bus-bar dead end structures shall be designed so as to be suitable for future bus-bar extension.

7.2 Standards

The steel structure shall conform to the following standards as to material, design and tests:

ISO 630	Structural Steels – Plates, wide flats, bars, sections and profiles
ISO 898	Mechanical Properties of Fasteners
JIS G3101	Rolled Steel for General Structure
JEC – 127	Design Standard for Transmission Equipment Support Structure
BS 729	Hot dip galvanized coatings on iron and steel
SNI No. 04-6918	Standar Nasional Indonesia
DIN 267	Mechanical characteristics of nuts
DIN 17100	Steels for General Structural Purposes

And/or other equivalent or better relevant standards shall be applied.

7.3 Design

- 7.3.1 All structures shall be designed so that no failure or permanent distortion shall occur when tested with applied forces equal to 2.5 times the maximum simultaneous working loads.
- 7.3.2 The maximum allowable stresses in tensile members shall be such as to give a factor of safety of not less than 2.5 on the elastic limit strength.
- 7.3.3 The following criteria shall be used for detailed design of the new respectively checking of the existing structures.

Vertical loading:

The dead weight of all conductors, earth-wires, insulator strings and apparatus supported by the structures and the structures themselves shall be considered.

Wind loading:

The wind load on the projected area shall include of the steel members, conductors, insulators and fittings as specified. The steelwork shall be designed to withstand simultaneously wind loads as follows:

	Wind pressure (daN/m ²)	Wind pressure coefficient	Resulting for design (daN/m ²)
Lattice steel Structure (towers, gantries)	70	2.8	196
Flat Surfaces	70	1.6	112
Conductors, insulators	53	1.0	53

Horizontal loading:

The pulls of phase conductors and earth-wires, taking into account the expected dynamic loads dictated by the specified short circuit current.

Details and fastenings shall be designed to have factor of safety against failure not less than the main members of the structure.

Short circuit loading (Initial symmetrical short-circuit current)

Stress diagrams and calculations shall be submitted as required by the Owner and the dispositions and sections of all members and the design of joints and fittings shall be subjected to approval.

- 7.3.4 Bidder should design the steel structure considering the applied conductor tension and sag for the condition, that at the highest operating temperature of conductor
- 7.3.5 Bolts and nuts shall be fitted with spring washers. Taper washers are to be added where necessary. Threads of bolts shall be spun galvanized and the threads of nuts shall be greased. The diameter of bolts and nuts which are

mechanically stressed shall not be less than 12 mm and shall have metric screw threads. Nuts and heads of all bolts shall be of the hexagonal type. Minimum quality of bolt shall be in accordance to the respective standards (5.6 of DIN 267 or equivalent standards or better).

Bolt holes are not to be more than 1.5 mm larger in diameter than the corresponding bolt diameter. The design is to be such as to keep the number of different parts as small as possible and is to facilitate transport, erection and inspection.

7.4 Material

7.4.1 The whole of the rolled steel sections, tubes, flats, plates, bolts, nuts and bars used shall employ weldable structural steel of an approved quality to ISO/R630. Rolled steel section, flats and plates used shall not be inferior in strength and quality to those specified as St 37 and St 52 in DIN 17100.

7.4.2 The steel shall be free from blisters, scale and other defects.

Main members and bracings of lattice structures shall be not less than 6 mm and 5 mm thick respectively.

7.5 Construction

7.5.1 The compression members of steel structures shall consists of rolled steel sections and the tension members of roiled steel sections or flats.

7.5.2 All members shall be stamped or marked for erection purposes as specified. All members shall be stamped or marked in an approved manner with numbers and/or letters corresponding to number and/or letter on the drawings or material lists. Drawings and material lists shall be submitted to the Owner for approval. The erection marks if stamped, shall be stamped before galvanizing and shall be clearly visible after galvanizing.

7.5.3 Pockets and depressions likely to hold water shall be avoided and all parts of the structures shall be properly drained.

7.5.4 Special care shall be taken not to injure the skin on galvanized or special treated surface during erection. Care shall be taken to prevent or remove any white rust, streaks or foreign matter deposited on galvanized surfaces during storing or transport or after erection.

- 7.5.5 Approved means shall be provided for fixing and bonding copper conductors to the steel work at sufficient points to obtain efficient earthing. Earth connection shall be made to a vertical face, clear of the ground. Foundation bolts shall not be used for their attachment.

7.6 Workmanship

- 7.6.1 All members shall be cut to jig and holes shall be drilled or punched to jig. All parts shall be carefully cut and holes accurately located so that when the members are in position the holes can be accurately aligned before being bolted up. Drifting or reaming of holes will not be permitted. All burrs shall be removed before galvanizing.

- 7.6.2 The drilling, punching, cutting, bending and welding of all fabricated steel-work shall be carried out before galvanizing and shall be such as to prevent possibility of irregularity occurring which might cause any difficulties in the erection of steel structures on the site.

Except where specified to the contract, all iron and steel used in the construction of the work shall be galvanized.

- 7.6.3 Galvanizing shall be applied by the hot dip process for all parts according to the relevant standard as mentioned under Section 7.2.

The zinc coating shall be smooth, clean and of uniform thickness and free from defects. The preparation for galvanizing itself shall not adversely affect the mechanical properties of the coated materials.

- 7.6.4 Built members shall, when finished, be true and free from all kinks, twists and open joints, and the material shall not be defective or strained in any way.

Steel gauges of the stud type shall be provided to enable the Owner to carry out such checking of members.

7.7 Other Materials

- 7.7.1 General

All construction materials such as conduit, steel angles, steel channels, steel plate, bolts, nuts and other related items required for operation shall be pro-

vided without extra charge and shall comply with the highest grade requirements of relevant standards.

7.8 Tests

The following tests shall be carried out before shipment

- 1) Mechanical strength of materials
- 2) Galvanizing test
- 3) Shop assembly

8 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

8.1 Quality assurance/quality control shall conform to the requirements of Part 4, Subsections 2.7, “Quality Assurance/Quality Control”.

8.2 The quality control procedures so specified shall include but not be limited to the following:

- Material certificates
- Welding
- Nondestructive testing
- Shop test and inspection procedures
- Cleaning and painting
- Packaging and shipping

8.3 The quality control documentation as specified above shall include but not be limited to the following:

- Material certificates
- Welding procedures
- Shop test and inspection reports

9 INSPECTION AND TEST AT SHOP

The Bidder shall submit a work inspection and tests program covering all materials and equipment to the Owner.

Agreement is to be established on the degree of inspection and test to be witnessed by the Owner or his representative before start and during the course of work.

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